

AgentQA — Exhaustive File-by-File Technical Blueprint & Master Documentation

This document contains the complete, low-level technical specification and line-by-line file scope for **all 22 files** in the **AgentQA** repository (`c:\Users\ADMIN\Desktop\Voziq\AgentQA`).

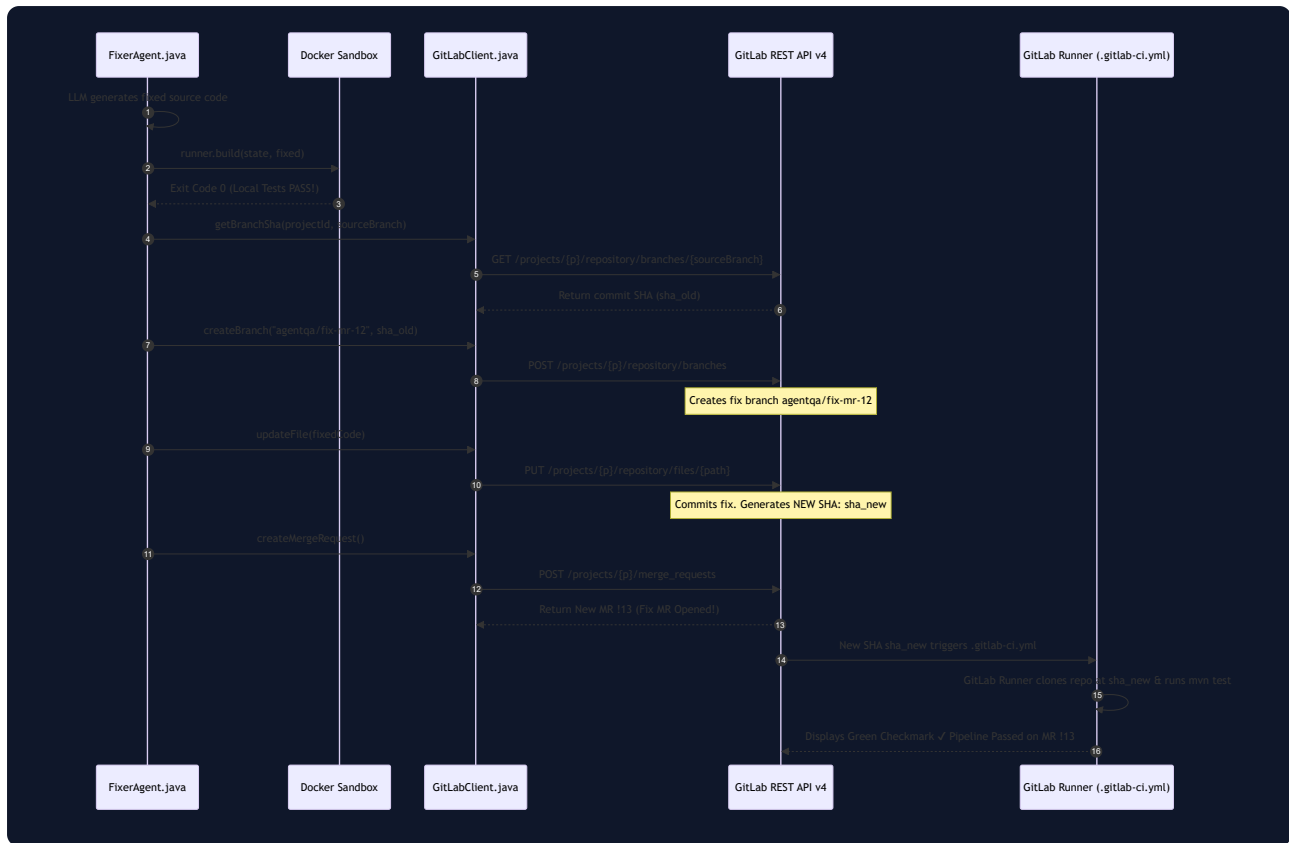
1. System Overview & Technology Matrix

AgentQA is an autonomous AI-driven QA system and self-healing MR pipeline built for GitLab with Java 21, Spring Boot 4.1.1, PostgreSQL 16, Groq LLM API, and Docker.

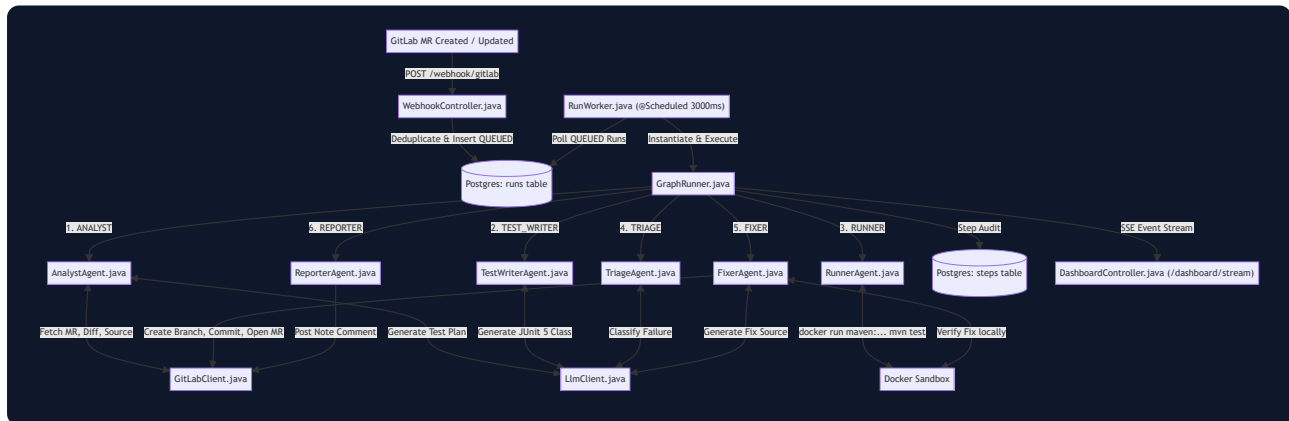
Layer	Technology	Version	Purpose in AgentQA
Language	Java 21	OpenJDK 21 / Temurin	Modern Java 21 features, record classes, pattern matching.
Framework	Spring Boot	v4.1.1	Application container, bean management, scheduling.
Web MVC	<code>spring-boot-starter-webmvc</code>	Servlet MVC	RestController for webhooks, REST endpoints, and SSE emitters.
HTTP Client	Spring <code>RestClient</code>	Synchronous RestClient	Communicates with Groq API and GitLab REST API v4.
JSON Provider	Jackson 3	<code>tools.jackson.databind</code>	JSON tree parsing for Spring Boot 4.
Persistence	Spring Data JPA + Hibernate	<code>ddl-auto=update</code>	ORM database mapping for <code>runs</code> and <code>steps</code> tables.
Database	PostgreSQL	v16 in Docker (Port 5433)	Task queue and step execution audit log storage.
Scheduling	<code>@EnableScheduling</code> + <code>@Scheduled</code>	<code>fixedDelay = 3000</code>	Asynchronous worker polling loop.
LLM Inference	Groq API	Model <code>openai/gpt-oss-120b</code>	High-speed LLM inference (2000 max tokens).
VCS Integration	GitLab REST API v4	Base URL: <code>https://gitlab.com</code>	API calls for fetching code, branches, commits, MR creation, notes.
Test Sandbox	Docker	<code>maven:3.9-eclipse-temurin-21</code>	Sandboxed container running JUnit 5.10.2 & Surefire 3.2.5.

FEATURE HIGHLIGHT: Fixer Agent to GitLab New Auto-Fix MR Creation Sequence Diagram

The sequence diagram below illustrates the exact step-by-step API flow when **FixerAgent** generates a code fix, re-verifies it in Docker locally, creates a branch, commits the fix, opens a **New Auto-Fix MR on GitLab**, and triggers `.gitlab-ci.yml` and **GitLab Runner**:



2. Master System Communications Flow



3. Line-by-Line Exhaustive File Scope Breakdown (All 22 Files)

A. Infrastructure & Build Files

1. pom.xml

- **File Location:** [pom.xml](#)
- **What it Created / Type:** Maven Project Object Model (Build Definition).

- **Purpose:** Defines dependencies, compiler plugins, and Java 21 build properties.
- **What Code it Includes:**
 - `java.version : 21`
 - `spring-boot-starter-parent : 4.1.1`
 - `spring-boot-starter-webmvc` : Web MVC & RestControllers.
 - `spring-boot-starter-data-jpa` : Spring Data JPA & Hibernate.
 - `postgresql` : JDBC Driver for PostgreSQL 16.
 - `spring-boot-devtools` : Hot reload tooling.
- **How it Connects:** Read by Maven (`mvnw` / `mvn`) to build and run the Spring Boot application.

2. `docker-compose.yml`

- **File Location:** [docker-compose.yml](#)
- **What it Created / Type:** Docker Compose Service Definition.
- **Purpose:** Manages the persistent PostgreSQL 16 database container.
- **What Specifications it Includes:**
 - Service `postgres` , container name `agentqa-postgres` .
 - Image `postgres:16` .
 - Port mapping `5433:5432` (Exposes PostgreSQL on host port 5433).
 - Environment variables: `POSTGRES_DB=agentqa` , `POSTGRES_USER=agentqa` , `POSTGRES_PASSWORD=agentqa` .
 - Volume mount `pgdata:/var/lib/postgresql/data` .
- **How it Connects:** Connects to Spring Boot via `spring.datasource.url` on port 5433.

3. `application.properties`

- **File Location:** [application.properties](#)
- **What it Created / Type:** Spring Application Configuration.
- **Purpose:** Stores runtime database credentials and API configuration properties.
- **What Properties it Includes:**
 - `spring.datasource.url : jdbc:postgresql://localhost:5433/agentqa`
 - `spring.jpa.hibernate.ddl-auto : update`
 - `agentqa.webhook.secret : dev-secret-123`
 - `agentqa.gitlab.base-url : https://gitlab.com`
 - `agentqa.gitlab.project-id : 85861338`
- **How it Connects:** Injected into `WebhookController` , `GitLabClient` , and `RunWorker` .

4. `AgentqaApplication.java`

- **File Location:** [AgentqaApplication.java](#)
- **What it Created / Type:** Java Class (`com.agentqa.AgentqaApplication`).
- **Purpose:** Main Spring Boot entry point.
- **What Code it Includes:** `@SpringBootApplication` , `@EnableScheduling` , `public static void main(String[] args)` .
- **How it Connects:** Boots Spring ApplicationContext, scans components, and enables `@Scheduled` worker tasks in `RunWorker.java` .

B. Core State Machine & Agent Framework

5. `Agent.java`

- **File Location:** [Agent.java](#)
- **What it Created / Type:** Java Interface + Inner Enum `Node` + Inner Class `State` .
- **Purpose:** Standardizes agent contract and defines the in-memory blackboard object carried across the graph.
- **What Code & Fields it Includes:**
 - Enum `Node` : `ANALYST` , `TEST_WRITER` , `RUNNER` , `TRIAGE` , `FIXER` , `REPORTER` , `END` .
 - Interface method: `Node node()` , `void execute(State state)` throws `Exception` .
 - Class `State` Fields: `runId` , `projectId` , `mrId` , `sourceBranch` , `headSha` , `testPlan` , `sourceFileName` , `sourceCode` , `generatedFiles` , `lastBuildError` , `writerAttempts` , `buildPassed` , `mrTitle` , `mrDescription` , `verdict` , `finding` , `fixLine` , `fixedCode` , `sourceFilePath` , `fixMrId` , `packageName` .
- **How it Connects:** Implemented by all 6 agents. Carried by `GraphRunner.java` .

6. GraphRunner.java

- **File Location:** [GraphRunner.java](#)
- **What it Created / Type:** Spring `@Component` Class (`com.agentqa.graph.GraphRunner`).
- **Purpose:** The hand-rolled state machine engine. Drives state transitions and publishes SSE events.
- **What Code it Includes:**
- Injected dependencies: `List<Agent>`, `StepRecordRepository`, `DashboardController`, `ObjectMapper`, `LlmClient`.
- Constructor maps agents: `agents = agentList.stream().collect(Collectors.toMap(Agent::node, Function.identity()))`.
- State machine loop (`run(state)`): loops while `node != Agent.Node.END`.
- Transition matrix (`next(current, state)`):
 - `ANALYST → TEST_WRITER`
 - `TEST_WRITER → RUNNER`
 - `RUNNER → (buildPassed ? REPORTER : TRIAGE)`
 - `TRIAGE → ("TEST_ERROR".equals(verdict) ? (writerAttempts < 3 ? TEST_WRITER : REPORTER) : FIXER)`
 - `FIXER → REPORTER`
 - `REPORTER → END`
- Real-time SSE publisher: `publish(state, agent, attempt, stateName, detail, ms)` $\rightarrow$ calls `dashboard.publish(json)`.
- **How it Connects:** Invoked by `RunWorker.java`. Controls agent execution order and logs audit records in `StepRecordRepository`.

C. The 6 Autonomous Agent Implementations

7. AnalystAgent.java

- **File Location:** [AnalystAgent.java](#)
- **What it Created / Type:** Spring `@Component` Class implementing `Agent` (`Agent.Node.ANALYST`).
- **Purpose:** Requirements analyst agent. Fetches MR info, extracts package declarations, and prompts LLM for JSON test plan.
- **What Code it Includes:**
- Dependencies: `LlmClient`, `GitLabClient`.
- Helper method `packageOf(source)`: Parses `package com.demo.shop;` header from source code.
- Inverted Prompt: *"Judge the code against what the merge request promises, not against itself."*
- Populates `state.setTestPlan()`, `state.setSourceFileName()`, `state.setSourceCode()`, `state.setPackageName()`.
- **How it Connects:** Invoked by `GraphRunner` at Node 1.

8. TestWriterAgent.java

- **File Location:** [TestWriterAgent.java](#)
- **What it Created / Type:** Spring `@Component` Class implementing `Agent` (`Agent.Node.TEST_WRITER`).
- **Purpose:** Generates JUnit 5 test classes.
- **What Code it Includes:**
- Dependencies: `LlmClient`.
- System Prompt Rules: Declares SAME package as class under test, no reflection, no assumptions, no stubbing, never weaken assertions.
- Retry logic: Appends `lastBuildError` when `state.getLastBuildError() != null`.
- Stores generated code in `state.getGeneratedFiles().add(code)`.
- **How it Connects:** Invoked by `GraphRunner` at Node 2.

9. RunnerAgent.java

- **File Location:** [RunnerAgent.java](#)
- **What it Created / Type:** Spring `@Component` Class implementing `Agent` (`Agent.Node.RUNNER`).
- **Purpose:** Executes tests inside isolated Docker sandbox container.
- **What Code it Includes:**
- `buildIn(work, state, sourceCode)`: Creates package subdirectories (`src/main/java/com/demo/shop/CartService.java` and `src/test/java/com/demo/shop/CartServiceTest.java`).
- Writes synthetic `pom.xml` (Java 21, JUnit 5.10.2, Surefire 3.2.5).

- Executes ProcessBuilder: `docker run --rm -v work:/app -v ~/.m2:/root/.m2 maven:3.9-eclipse-temurin-21 mvn -q -B test`.
- Timeout: 5 minutes (`process.waitFor(5, TimeUnit.MINUTES)`).
- Exit code 0 $\rightarrow$ `buildPassed = true` ; Exit code != 0 $\rightarrow$ `buildPassed = false` , trims log to 60 lines.
- **How it Connects:** Invoked by `GraphRunner` at Node 3. Also invoked by `FixerAgent` for local re-verification!

10. TriageAgent.java

- **File Location:** [TriageAgent.java](#)
- **What it Created / Type:** Spring `@Component` Class implementing `Agent` (`Agent.Node.TRIAGE`).
- **Purpose:** Failure classification engineer.
- **What Code it Includes:**
- Dependencies: `LlmClient` .
- System prompt: Returns JSON `{"verdict":"TEST_ERROR|CODE_DEFECT", "finding":" ... "}` .
- Verdict parsing: `reply.contains("CODE_DEFECT")` $\rightarrow$ `verdict = "CODE_DEFECT"` , else `"TEST_ERROR"` .
- **How it Connects:** Invoked by `GraphRunner` at Node 4 on build failure.

11. FixerAgent.java

- **File Location:** [FixerAgent.java](#)
- **What it Created / Type:** Spring `@Component` Class implementing `Agent` (`Agent.Node.FIXER`).
- **Purpose:** Autonomous code repair engineer & Auto-Fix MR author.
- **What Code it Includes:**
- Dependencies: `LlmClient` , `GitLabClient` , `RunnerAgent` .
- System prompt: *"Return the COMPLETE corrected source file including its package declaration..."*
- Re-verification: Calls `runner.build(state, fixed)` locally in Docker (aborts if `fixWorks == false`).
- Auto-Fix MR sequence:
 1. `gitlab.getBranchSha()`
 2. `gitlab.createBranch(projectId, "agentqa/fix-mr-" + mrId, headSha)`
 3. `gitlab.updateFile(projectId, sourceFilePath, branch, fixed, message)`
 4. `gitlab.createMergeRequest(projectId, branch, sourceBranch, title, description)`
- Stores `state.setFixMrId(id)` .
- **How it Connects:** Invoked by `GraphRunner` at Node 5 when `verdict == "CODE_DEFECT"` .

12. ReporterAgent.java

- **File Location:** [ReporterAgent.java](#)
- **What it Created / Type:** Spring `@Component` Class implementing `Agent` (`Agent.Node.REPORTER`).
- **Purpose:** Report communicator.
- **What Code it Includes:**
- Dependencies: `GitLabClient` .
- Formats Markdown report (Passed / Defect Found + Fix MR Link / Retries Exhausted).
- Calls `gitlab.postComment(projectId, mrId, body)` .
- **How it Connects:** Invoked by `GraphRunner` at Node 6.

D. External Client Integration Services

13. GitLabClient.java

- **File Location:** [GitLabClient.java](#)
- **What it Created / Type:** Spring `@Component` Class (`com.agentqa.gitlab.GitLabClient`).
- **Purpose:** REST API v4 wrapper for GitLab.
- **What Code & Methods it Includes:**
- Injected property: `@Value("${agentqa.gitlab.base-url}") String baseUrl` . Validates `GITLAB_TOKEN` env var.
- Methods:
 - `getMergeRequest(projectId, mrId)` $\rightarrow$ `GET /projects/{p}/merge_requests/{i}`
 - `getChanges(projectId, mrId)` $\rightarrow$ `GET /projects/{p}/merge_requests/{i}/changes`
 - `getFile(projectId, path, ref)` $\rightarrow$ `GET /projects/{p}/repository/files/{path}/raw`
 - `postComment(projectId, mrId, body)` $\rightarrow$ `POST /projects/{p}/merge_requests/{i}/notes`
 - `getBranchSha(projectId, branch)` $\rightarrow$ `GET /projects/{p}/repository/branches/{branch}`
 - `createBranch(projectId, newBranch, fromRef)` $\rightarrow$ `POST /projects/{p}/repository/branches`

- `updateFile(projectId, path, branch, content, message)` $\rightarrow$ PUT `/projects/{p}/repository/files/{path}`
- `createMergeRequest(projectId, source, target, title, description)` $\rightarrow$ POST `/projects/{p}/merge_requests`
- **How it Connects:** Called by `AnalystAgent`, `FixerAgent`, and `ReporterAgent`.

14. `LlmClient.java`

- **File Location:** [LlmClient.java](#)
- **What it Created / Type:** Spring `@Component` Class (`com.agentqa.llm.LlmClient`).
- **Purpose:** REST API client for Groq LLM inference.
- **What Code & Methods it Includes:**
- Validates `GROQ_API_KEY` env var.
- Base URL: `https://api.groq.com`, Model: `openai/gpt-oss-120b`, Max tokens: 2000.
- Method `complete(systemPrompt, userPrompt)` $\rightarrow$ POST `/openai/v1/chat/completions`.
- Method `field(json, key, default)` $\rightarrow$ Helper to extract JSON fields safely.
- **How it Connects:** Called by `AnalystAgent`, `TestWriterAgent`, `TriageAgent`, `FixerAgent`, and `GraphRunner`.

E. Webhook Ingress & Task Worker

15. `WebhookController.java`

- **File Location:** [WebhookController.java](#)
- **What it Created / Type:** Spring `@RestController` Class (`com.agentqa.webhook.WebhookController`).
- **Purpose:** Ingress endpoint for GitLab webhooks.
- **What Code it Includes:**
- Injected dependencies: `RunRepository`, `@Value("${agentqa.webhook.secret}") String secret`.
- Endpoint: `@PostMapping("/gitlab")`.
- Checks header `X-Gitlab-Token` matches `secret`.
- Filters `object_kind == "merge_request"` and action in `["open", "update", "reopen"]`.
- Deduplicates via `runRepository.existsByProjectIdAndMrIdAndHeadShaAndStatusIn(QUEUED, RUNNING)`.
- Saves new `Run(QUEUED)` entity. Returns `200 OK ("queued run {id}")`.
- **How it Connects:** Receives HTTP requests from GitLab Webhooks. Saves runs to PostgreSQL `runs` table.

16. `RunWorker.java`

- **File Location:** [RunWorker.java](#)
- **What it Created / Type:** Spring `@Component` Class (`com.agentqa.worker.RunWorker`).
- **Purpose:** Background queue polling worker.
- **What Code it Includes:**
- Injected dependencies: `RunRepository`, `GraphRunner`.
- Scheduled method: `@Scheduled(fixedDelay = 3000) public void pollQueue()`.
- Queries `runRepository.findByStatusOrderByCreatedAtAsc(RunStatus.QUEUED)`.
- Updates run status to `RUNNING`.
- Instantiates `Agent.State state = new Agent.State(...)`.
- Calls `graphRunner.run(state)`.
- Catches exceptions: sets status to `DONE` on success or `FAILED` on exception.
- **How it Connects:** Scheduled by Spring framework every 3 seconds. Triggers `GraphRunner`.

F. Database Entities & Repositories

17. `Run.java` & `RunStatus.java`

- **File Location:** [Run.java](#), [RunStatus.java](#)
- **What it Created / Type:** JPA `@Entity` Class + Enum `RunStatus`.
- **Purpose:** Maps to PostgreSQL `runs` table.
- **What Code it Includes:**
- `@Table(name = "runs")`
- Fields: `id` (PK, `@GeneratedValue`), `projectId`, `mrId`, `sourceBranch`, `headSha`, `status` (Enum `QUEUED`, `RUNNING`, `WAITING_APPROVAL`, `DONE`, `FAILED`, `REJECTED`), `createdAt`, `updatedAt`.

- **How it Connects:** Managed by Hibernate / JPA. Queried by `RunRepository` .

18. `RunRepository.java`

- **File Location:** `RunRepository.java`
- **What it Created / Type:** Spring Data JPA Interface extending `JpaRepository<Run, Long>` .
- **Purpose:** Database access interface for `Run` entities.
- **What Query Methods it Includes:**
- `List<Run> findByStatusOrderByCreatedAtAsc(RunStatus status)`
- `Optional<Run> findFirstByProjectIdAndMrIdAndStatus( ... )`
- `boolean existsByProjectIdAndMrIdAndHeadShaAndStatusIn( ... )`
- **How it Connects:** Used by `WebhookController` and `RunWorker` .

19. `StepRecord.java` & `StepRecordRepository.java`

- **File Location:** `StepRecord.java`, `StepRecordRepository.java`
- **What it Created / Type:** JPA `@Entity` Class + `JpaRepository` Interface.
- **Purpose:** Maps to PostgreSQL `steps` table for audit trail.
- **What Code it Includes:**
- `@Table(name = "steps")`
- Fields: `id`, `runId` (FK to `runs.id`), `agent`, `attempt`, `startedAt`, `finishedAt`, `outcome` ("OK", "FAILED"), `detail` .
- Repository method: `findByRunIdOrderByStartedAtAsc(Long runId)` .
- **How it Connects:** Used by `GraphRunner` to record audit steps and `RunController` to display execution paths.

G. Monitoring & Dashboard Web Controllers

20. `RunController.java`

- **File Location:** `RunController.java`
- **What it Created / Type:** Spring `@RestController` Class (`com.agentqa.run.RunController`).
- **Purpose:** REST API endpoints for querying run timelines and step audit history.
- **What Code & Endpoints it Includes:**
- Injected dependencies: `RunRepository`, `StepRecordRepository` .
- `@GetMapping("/{id}")` : Returns `RunDetail` JSON containing full step path (e.g. `ANALYST#1 → TEST_WRITER#1 → RUNNER#1 → TRIAGE#1 → FIXER#1 → REPORTER#1`) and step durations.
- `@GetMapping` : Returns `List<RunSummary>` recent runs.
- **How it Connects:** Called by web clients / CLI for status auditing.

21. `DashboardController.java`

- **File Location:** `DashboardController.java`
- **What it Created / Type:** Spring `@RestController` Class (`com.agentqa.dashboard.DashboardController`).
- **Purpose:** Server-Sent Events (SSE) streaming controller for real-time live UI updates.
- **What Code it Includes:**
- `CopyOnWriteArrayList<SseEmitter> clients` .
- `@GetMapping("/dashboard/stream")` : Registers new client SSE stream.
- `publish(String json)` : Broadcasts real-time JSON events to all connected clients.
- **How it Connects:** `GraphRunner` calls `dashboard.publish()` on every step. Streams live events to `dashboard.html` .

22. `dashboard.html`

- **File Location:** `dashboard.html`
- **What it Created / Type:** Static HTML/JS Web Dashboard UI.
- **Purpose:** Real-time web user interface for watching AgentQA executions.
- **What Code it Includes:**
- JavaScript `EventSource('/dashboard/stream')` subscriber.
- Displays live timeline cards, risk levels, build pass/fail badges, defect findings, and direct links to newly opened Auto-Fix MRs (`fixMr`)!
- **How it Connects:** Served by Spring Boot static asset handler at `http://localhost:8080/dashboard.html` .

4. Summary Matrix of File Interconnections

Source File	Primary Dependents (Callers / Users)	Downstream Dependencies Called	External Systems Contacted
AgentqaApplication	Spring Framework Bootstrapper	RunWorker	None
WebhookController	GitLab Webhook Ingress	RunRepository , Run	GitLab Webhook Client
RunWorker	@Scheduled Spring Thread	RunRepository , GraphRunner , Agent.State	PostgreSQL Database
GraphRunner	RunWorker	All 6 Agent beans, StepRecordRepository , DashboardController , LlmClient	PostgreSQL Database
AnalystAgent	GraphRunner	GitLabClient , LlmClient	GitLab REST API, Groq LLM API
TestWriterAgent	GraphRunner	LlmClient	Groq LLM API
RunnerAgent	GraphRunner , FixerAgent	Native ProcessBuilder	Docker Sandbox (maven:3.9-eclipse-temurin-21)
TriageAgent	GraphRunner	LlmClient	Groq LLM API
FixerAgent	GraphRunner	LlmClient , RunnerAgent , GitLabClient	Groq LLM API, Docker Sandbox, GitLab REST API
ReporterAgent	GraphRunner	GitLabClient	GitLab REST API
GitLabClient	AnalystAgent , FixerAgent , ReporterAgent	Spring RestClient	GitLab REST API v4 (https://gitlab.com)
LlmClient	AnalystAgent , TestWriterAgent , TriageAgent , FixerAgent , GraphRunner	Spring RestClient	Groq LLM API (https://api.groq.com)
DashboardController	GraphRunner	SseEmitter	Browser EventSource (dashboard.html)
RunController	Browser / REST Clients	RunRepository , StepRecordRepository	PostgreSQL Database